

Lab 1 — Requirements Specification (SRS Extract)

Restaurant Table Booking & Pre-Ordering App | Problem Statement #36 | Retail, E-Commerce & Finance

PES University — Dept. of CSE • Requirements Engineering & UML Use-Case Modelling • Name: _____ SRN: _____ Date: _____

1. Scenario Summary

A restaurant hospitality app that lets diners reserve a specific table from a live 2D floor plan, place their food order in advance, and have the kitchen time preparation so that the meal is ready when they arrive. Restaurant Managers work the same floor plan from in-house terminals, moving tables through the Reserved → Seated → Billed lifecycle.

Actors: Diner (primary), Restaurant Manager (primary), Payment Gateway (secondary/external), Kitchen Display System — KDS (secondary/external).

2. Requirements Table (5 FRs + 2 NFRs)

Req ID	Type	Description ("The system shall...")	Priority	Acceptance Criteria (measurable pass/fail)	Rationale (short)	Comments (peer critique)
FR-001	Functional	The system shall allow a Diner to select an available table from the restaurant's interactive 2D floor plan for a chosen date, time slot and party size, and shall link the diner's pre-ordered menu items to that reservation.	High	Pass: tapping a table shown as "Available" and confirming creates one reservation record with a unique booking ID, locks that table for the entire dining window (slot + 90 min) and queues a kitchen pre-order ticket against the same booking ID. Fail: any overlapping reservation is accepted for the same table — 0 double bookings permitted in 100 concurrent booking attempts.	Core ordering functionality of the app; table locking is what prevents overbooking.	Define the dining-window length per table size in the data dictionary.
FR-002	Functional	The system shall allow a Diner to browse the live digital menu and add items — with quantity and customisation notes — to a pre-order cart attached to the pending reservation.	High	Pass: each added item appears in the cart with the correct line total and running total, and is visible against the booking ID on the Restaurant Manager's terminal within 2 s. Removing an item recalculates the total in the same screen refresh. Fail: cart total ≠ sum of line totals, or item not visible to the manager.	Pre-ordering is the differentiating feature; the kitchen cannot plan prep without the item list.	State behaviour for items that go out of stock after being added.
FR-003	Functional	The system shall calculate a kitchen preparation timeline for every pre-order by deriving each item's cook-start time from its standard prep duration and the reserved arrival time, and shall release the ticket to the Kitchen Display System at that cook-start time.	High	Pass: for a 19:30 reservation containing a 10-minute item and a 25-minute item, the KDS receives the tickets at 19:20 and 19:05 (± 60 s) and the dispatch timestamps are written to the audit log. Fail: any ticket released more than 60 s early or late, or not released at all.	Delivers the "seamless arrival dining" goal — food is ready as the diner is seated.	Prep durations should be maintained by the manager, not hard-coded.
FR-004	Functional	The system shall allow the Diner to pay the pre-order amount (or booking deposit) through the integrated payment gateway and shall issue a digital confirmation containing the booking ID, table number, itemised pre-order and a QR check-in code.	High	Pass: on an "authorised" response the reservation status changes to CONFIRMED and the confirmation is delivered by push and e-mail within 30 s. Fail: on decline the reservation remains PENDING, no ticket is queued, and the table hold is released after 2 failed attempts.	Payment commits the diner to the booking and reduces no-shows and wasted prep.	Name the gateway (e.g. Razorpay) and the authorisation timeout value.
FR-005	Functional	The system shall allow the Restaurant Manager to update a table's lifecycle status (Available → Reserved → Seated → Billed) from the floor terminal, and shall automatically return a reserved table to "Available" if the diner is not marked "Seated" within 15 minutes of the slot start time.	Medium	Pass: a status change made on one terminal is reflected on every other terminal and on the diner floor plan; a table whose booking is not marked Seated is released at slot start + 15 min and the reservation is flagged NO_SHOW. Fail: a released table remains blocked, or a NO_SHOW booking still holds the table.	Keeps the live floor plan truthful and recovers covers lost to no-shows during peak service.	Make the 15-minute grace period configurable per restaurant.

Req ID	Type	Description ("The system shall...")	Priority	Acceptance Criteria (measurable pass/fail)	Rationale (short)	Comments (peer critique)
NFR-001	Non-functional (Performance)	The system shall propagate any table status change (Reserved, Seated, Billed, Available) to all manager floor terminals and to the diner-facing floor plan in under 500 ms.	High	Pass: a k6/JMeter benchmark at peak load (200 concurrent sessions, 50 status changes per minute) reports p95 ≤ 500 ms and p99 ≤ 800 ms end-to-end, with no stale table state after the run. Fail: p95 > 500 ms or any terminal showing stale status.	Stale floor state causes double seating and overbooking during peak dining hours.	<i>Specify the measurement point — server ack vs. terminal repaint.</i>
NFR-002	Non-functional (Security)	The system shall never persist raw cardholder data: card details shall be captured directly by the PCI-DSS compliant payment gateway and retained only as a token, and all traffic between the app, backend and gateway shall use TLS 1.2 or higher.	High	Pass: an automated scan of the database, application logs and backups finds no PAN or CVV; the public endpoint scores A on an SSL/TLS configuration test; a pen-test report shows no card data in transit outside TLS. Fail: any card data found at rest or in logs.	PCI DSS obligation — payment data is the highest-value asset in the system.	<i>Add token-rotation / gateway key-management to the design notes.</i>

Priority scale: High = release blocker, Medium = required for launch but can follow the first increment, Low = desirable. Every acceptance criterion above is stated as a measurable pass/fail condition.

3. Actors, Use Cases & Traceability

UC ID	Use Case	Actor(s)	Traces to	Relationship
UC-01	Book Table	Diner	FR-001	«include» UC-02, «include» UC-05; extended by UC-03
UC-02	Validate Table Availability	— (included)	FR-001	Included by UC-01
UC-03	Pre-Order Menu Items	Diner	FR-002	«extend» UC-01 (optional); «include» UC-04
UC-04	Generate Kitchen Prep Timeline	Kitchen Display System	FR-003	Included by UC-03
UC-05	Make Payment	Payment Gateway	FR-004	Included by UC-01; extended by UC-06
UC-06	Apply Promo Code	— (extension)	FR-004	«extend» UC-05 (conditional)
UC-07	Manage Table Status	Restaurant Manager	FR-005, NFR-001	Association only
UC-08	Cancel / Release Reservation	Diner, Restaurant Manager	FR-005	Association only

All five functional requirements and both non-functional requirements are traced to at least one use case; no use case exists without a source requirement.